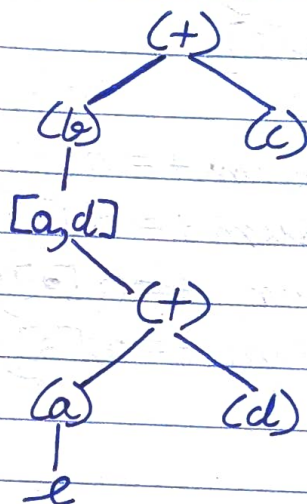


10) $a = b + c$
 $d = b + c$
 $e = a + d$

Expression $b + c$ appears twice, so it is a common subexpression.

DAG representation:-



Optimization:-

Since $a = b + c$

$d = b + c$

Compute once and reuse.

$t = b + c$

$a = t$

$d = t$

$e = t + t$

- Removed duplicate computation of $b + c$.
- Used temporary variable t .
- Reduced number of operations.